

SiO₂-supported HClO₄ catalyzed synthesis of (Z)-thiazolylhydrazonoindolin-2-ones and their electrochemical properties

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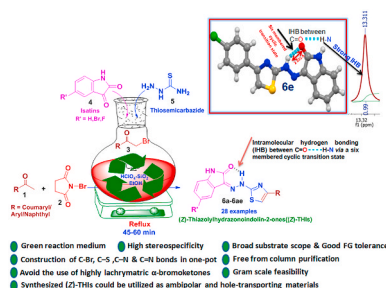
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HIGHLIGHTS

- Environmentally benign HClO₄-SiO₂ catalyzed FCDR strategy for synthesis of (Z)-THIs.
- Z-configuration of THIs confirmed by single crystal XRD and ¹H NMR studies.
- Construction of C-Br, C-N, C-S and C=N bonds in a single-step operation.
- (Z)-THIs could be promising ambipolar materials for efficient OLEDs.

GRAPHICAL ABSTRACT



ARTICLE INFO

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ABSTRACT

In this paper, an environmentally benign silica-supported perchloric acid (HClO₄-SiO₂) catalyzed green FCDR strategy has been developed for the synthesis of (Z)-THIs (6) with high stereospecificity via an intramolecular hydrogen bond (IHB) directed approach, involving the reaction of methyl ketones (1), N-bromosuccinimide (NBS) (2), isatins (4) and thiosemicarbazide (5) in ethanol at reflux temperature for 45–60 min in one-pot. The reaction proceeds through the construction of C-Br (α-bromination), C-S & C-N (heterocyclization), and C=N (condensation) bonds in one pot. The absolute structure of the compound (Z)-3-(2-(4-(4-bromophenyl)thiazol-2-yl)hydrazono)indolin-2-one (6e) has been confirmed by single-crystal XRD analysis. Further, the role of IHB on Z-configuration of the synthesized (Z)-THIs is proved by single-crystal XRD and ¹H NMR studies. Wide substrate scope, good functional group tolerance, scalability, improved safety since the method circumvents the use of highly lachrymatory α-bromoketones as starting materials, high product yields (up to 98%), short reaction times, reusable solid Brønsted acid catalyst (HClO₄-SiO₂), and products that do not require column chromatography

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purification are all attractive features of this FCDR strategy. Electrochemical properties of THIs (6) are examined by cyclic voltammetry. The HOMO and LUMO energy level of THIs, 6a, 6c, 6d, 6j, 6o-6v, 6y, and 6aa are comparable with the reported ambipolar materials, and the HOMO levels of other THIs, 6b, 6e-6i, 6n, 6w, 6x, 6z and 6 ab-6ae are similar with the most commonly used hole transporting materials (HTMs).

1. Introduction

Intramolecular hydrogen bonds (IHBs) occupy a unique place not only in the determination of yield of the cis–trans isomers but also in photophysics, photochemistry, and biology (Coimbra et al., 2021; Santos-Martins and Forli, 2020; Pedersen et al., 2014; Pace et al., 2014; Simon and Goodman, 2010; Cui et al., 2012; Shalaeva et al., 2013; Gopinath et al., 2019). The presence of IHB in a drug molecule can markedly alter the properties like stability, solubility, physicochemical parameters, protein binding affinity, etc (Kujawski et al., 2015). For instance, teriflunomide is utilized as a medicine for multiple sclerosis, and the (Z)-isomer of teriflunomide is more stable and significant isomer than E-isomer. This is due to the existence of an IHB in the Z-isomer (Alex et al., 2011; Shalaeva et al., 2013; Doak et al., 2014). Further, these HBs also control the reactivity and selectivity in various organic reactions like aldol, epoxidation, hydrogenation, cycloaddition, Michael, and [2 + 2] cyclo addition reactions (Adam and Wirth, 1999; Schreiner, 2003; Taylor and Jacobsen, 2006; Bassani et al., 2000).

In recent days, the development of simple, step-economical, and environmentally compatible synthetic routes has been of prime importance in view of environmental sustainability (Anastas and Warner, 2000; Dunn et al., 2010). Multi-component domino reactions (MCDRs) have emerged as powerful strategies in modern synthetic organic chemistry because of their green characteristics as well as rapid and easy access to diversity-oriented synthesis (DOS) of valuable heterocycles (Hayashi, 2021; Nariyoshi and Hayashi, 2020; Sun et al., 2019; Yang et al., 2017). Nitrogen and sulphur based small heterocycles are considered as an important class of organic semiconducting materials because of their promising applications in fabricating optoelectronic devices such as organic light emitting diodes (OLEDs), organic photovoltaics (OPVs), organic field effect transistors (OFETs), etc (Panneerselvam et al., 2017; Zhao et al., 2016; Huang et al., 2017).

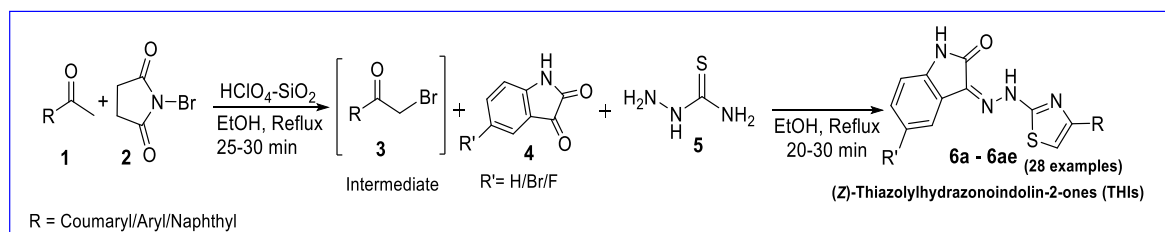
OLEDs are one of the leading technologies for flat-panel displays and energy-efficient lighting sources because of their unique features like excellent color tunability, high fluorescence efficiency, wide viewing angle, mercury-free manufacture, superior image quality, low-power consumption, low cost, flexibility/stretchability, etc (Kim et al., 2011; Zou et al., 2020). Recently, the most renowned electronics companies, including Apple, Samsung, LG, Google, Philips/OLED Works, Osram, and Konica Minolta offering OLED-based displays in their new products like cell phones, ultra-high-definition televisions, digital cameras, etc (Dubey et al., 2020; Nagar et al., 2022). An OLED device is a multi-layered structure that consists of a hole-injection/transport layer (HIL/HTL), emissive layer (EML), and electron-injection/transport layer (EIL/ETL), which are sandwiched by electrodes (Zhou et al., 2021; Amruth et al., 2021; Yadav et al., 2020). The chemical structure, morphology, thermal stability, charge mobility, and molecular orbital (highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO)) levels of each layer have played a key role in the long lifetimes and high power efficacy of OLEDs. The basic requirements for conventional hole transport materials (HTMs) are good solubility in non-polar solvents, suitable energy levels (HOMO and LUMO), appropriate glass transition temperature (T_g), and high hole mobility. Organic HTMs have a faster-growing research field than inorganic HTMs because the inorganic HTMs are deposited via a sol-gel process in polar solvents followed by annealing, which can damage the emissive layer (Kim et al., 2020).

Organic HTMs have several advantages over inorganic HTMs (NiO_x , CuI, Cu_xO , etc.) such as its good thin film-forming characteristics, less

toxic, low-cost, good solubility, environmental friendliness, solution-processability, flexibility and tunability, easy fabrication, etc. (Laila et al., 2021). From the literature, it is clear that the metal oxides, TiO_2 , SnO_2 , ZnO are employed as conventional electron transport materials (ETMs) in optoelectronic devices (Zheng et al., 2019; Jameel et al., 2021). These metal oxides suffer from disadvantages like high degradation under UV light, unsuitable for flexibility and tunability, high processing cost, hysteresis issues, etc. Recently, enormous research has been conducted on organic ETMs. The reports reveal the narrow performance gap between metal oxide-based ETMs and organic ETMs materials (Seitkhan et al., 2019; Wu et al., 2020). Further, organic ETMs have significant merits over inorganic metal oxide-based ETMs, which include easy preparation, lower processing cost, excellent solubility in non-polar solvents, mechanical flexibility, and broad range tunability (Zheng et al., 2019; Jameel et al., 2021). Ambipolar materials (APMs) have the unique property of transporting both holes and electrons at the same time under ambient conditions with the HOMO and LUMO energy levels of ≥ -5.6 eV and ~ -3.15 eV, respectively (Singh et al., 2019; Zhou et al., 2014; Shaikh and ChackoKamble, 2017).

The development of highly efficient organic APMs (Zhao et al., 2016; Singh et al., 2019; Prosa et al., 2020) for the fabrication of environmentally stable, cost-effective, and high-performance OLED devices is one of the stimulating areas of current research. The electro-analytical techniques received substantial attention owing to their operational simplicity, free from pretreatment procedures, high sensitivity, cost-effectiveness, the possibility of real-time results, efficiency in the determination of colours, etc. (Vladislavić et al., 2018; Bakirhan and Ozkan, 2018; Karimi-Maleh et al., 2022). Among the electro-analytical techniques, cyclic voltammetry (CV) is an excellent electrochemical method to calculate the HOMO and LUMO levels from onset oxidation and onset reduction potentials of respective organic materials. From this valuable information, highly efficient hole transport, electron transport, and ambipolar materials can be determined for the fabrication of highly efficient OLED devices (Yadav et al., 2020).

Thiazoles and isatins are considered as privileged N and S-based heterocycles, owing to their potential biological and optoelectronic properties. Thiazole derivatives occupy an important place in medicinal and materials chemistry due to their promising biological and optoelectronic applications (Aggarwal et al., 2013; Azizmohammadi et al., 2013; Siddiqui and Ahsan, 2010; Qin et al., 2014; Kowalska et al., 2019; Zeng et al., 2016; Kudrjasova et al., 2014; Hela et al., 2012; Giridhar et al., 2013). Besides, the thiazole ring system is the structural motif found in many drug molecules, for instance, Ritonavir, Bleomycin, Meloxicam, Fanetizole, and Tiazofurin (Izawa and Onishi, 2006; Lopes and Schwartzmann, 2001; Jayarama et al., 1986; Lednicer et al., 1990). Moreover, isatin (1H-Indole-2,3-dione) and its derivatives display various applications in biology and organic electronics (Medvedev et al., 2006; Andreani et al., 2010). These compounds are key structural units in several synthetic drugs like Sunitinib, Toceranib, and Nintedanib (Xiang et al., 2011; Griffith et al., 2006; Pandit et al., 2006). On the other hand, thiazolylhydrazonoindolin-2-ones (THIs) act as potent α -glucosidase inhibitors, anti-proliferative, antiviral agents, etc. (Singh et al., 1983; Rahim et al., 2015; Wang et al., 2016; Xie et al., 2017; Eldehna et al., 2018). To date, very limited reports are available on both step-wise and multi-component reaction strategies for the preparation of THIs. The reported step-wise procedure proceeds via thiosemicarbazone of isatin intermediate from the reaction between isatin and thiosemicarbazide and then reaction with phenacyl bromides (Kondratieva et al., 2007; Gao and Li, 2017). Tiwari et al. developed a 3-component reaction for



Scheme 1. HClO₄-SiO₂ catalyzed four component domino synthesis of (Z)-THIs (6).

the synthesis of THI derivatives from isatins, thiosemicarbazide, and phenacyl bromides catalyzed by CTAB in glycerol (Tiwari et al., 2018). Tewodros Birhanu et al. and Navaneeth Gowda et al. also reported a 3-component reaction for the synthesis of THI derivatives from isatins, thiosemicarbazide, and phenacyl bromides/3-(2-bromoacetyl)coumarin in the presence of acetic acid in ethanol under reflux conditions (Tewodros Birhanu and Rajeswar Rao, 2014; Navaneeth Gowda et al., 2022). In spite of this development, there are some shortcomings that include the use of highly lacrymating phenacyl bromides as one of the starting materials, limited substrate scope, long reaction times, etc. Hence, the development of an improved method that overcomes the above limitations is highly desirable for the synthesis of THIs from readily available starting materials, especially under green reaction conditions.

The placement of the homogeneous catalysts on surface active sites of heterogeneous catalyst supports is one of the important strategies in modern catalysis research. Catalyst supports play an important role in increasing the efficiency of the supported metal or metal oxide or Brønsted or Lewis acid catalysts on account of their chemical stability, high surface area, porosity, as well as capability of dispersing catalysts highly over their surface. A wide variety of catalyst supports for different catalysts for example, boehmite nanoparticles supported organic-inorganic hybrids (Mohammadi et al., 2021), Fe₃O₄ based magnetic nanoparticles (MNPs) supported various metal/organic catalysts (Kazemi and Mohammadi, 2019), SBA-15 supported guanine-lanthanum and cobalt complexes (Nikoorazm et al., 2020a,b; Ghorbani-Choghamarani et al., 2019), MCM-41 nanoparticles supported i) palladium complex [Pd(0)-guanidine@MCM-41] and ii) Zirconium@guanine nanoparticles (Filian et al., 2020; Nikoorazm et al., 2020a,b), mercynite MNPs supported i) sulfuric acid, ii) L-methionine-Pd complex and iii) guanidino L-arginine nickel complex (Mohammadi and Ghorbani-Choghamarani, 2022a,b; Mohammadi and Ghorbani-Choghamarani, 2020; Mohammadi and Ghorbani-Choghamarani, 2022a,b), SnFe₂O₄ MNPs supported silica sulfuric acid (Esmaili et al., 2022), UiO-66-NH₂(Zr) metal-organic Frameworks supported schiff-base-Cu(II) complex (Ghobakhloo et al., 2022a, 2022b), LDH modified γ-Fe₂O₃ MNPs supported bifunctional zwitter ionic HEPES (Ghobakhloo et al., 2022a, 2022b), halloysite and g-C₃N₄ supported dysprosium-balsalazide complex (Rahimzadeh et al., 2022) have been employed as heterogeneous catalysts in various organic transformations.

Further, other supported catalysts, copper/copper oxide nanoparticles (Cu/CuONPs) absorbed on various solid supports such as TiO₂ (CuONPs/TiO₂) (Trivikram Reddy et al., 2018), activated carbon (CuONPs/C) (Albaladejo et al., 2013), silica coated maghemite (CuNP-s/MagSilica) (Nador et al., 2013), zeolite (Alonso et al., 2015), graphene oxide (CuONPs/GO) (Hanuman Reddy et al., 2016), silica supported metal-based catalysts i) Fe/Fe-O nanoparticles (Chandrasekhar et al., 2022), ii) Au(I) complex (Shu et al., 2015), iii) Pd-HPW/SiO₂ (Palladium-doped silica-supported heteropoly acid) (Bakht et al., 2021) etc., silica supported Lewis acids i) Cu(OTf)₂ (Vijaya et al., 1999), ii) ZrCl₄ (Patney and Margan, 1996), iii) TaCl₅ (Chandrasekhar et al., 1997) etc., silica supported Brønsted acids i) 12-tungstophosphoric acid, ii) phosphomolybdic acid, iii) triflic acid, iv) polyphosphoric acid, v) fluoroboric acid, vi) sulfuric acid, vii) perchloric acid (Kaur et al., 2015)

have been employed as heterogeneous catalysts in various organic chemical transformations. Among the plethora of catalyst supports mentioned above, inexpensive and readily available silica (SiO₂) acts as excellent catalyst support because of its well-ordered porosity, high surface area, and good chemical stability. Silica-supported Brønsted acids have gained much attention in organic synthesis over the silica-supported metal-based catalysts and Lewis acids due to its unique advantages i.e. products free from hazardous and expensive metal contamination, ease of preparation and handling, high efficiency, low toxicity, increased selectivity, avoid the unwanted side reactions, etc. Motivated by our recent report on silica-supported perchloric acid (HClO₄-SiO₂) (Kumar et al., 2021), it is decided to study the present 4-component domino reaction (FCDR) strategy (Scheme 1) in the presence of this inexpensive and environmentally benign catalyst system.

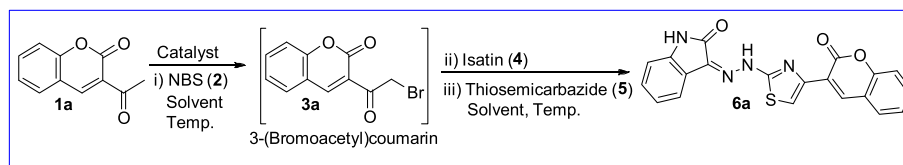
The present study aims to fulfill the following objectives; (i) develop a practical, green and sustainable FCDR strategy for (Z)-specific synthesis of THIs by the integration of bioactive isatin, thiazole, and imine pharmacophores using a molecular hybridization approach (Viegas-Junior et al., 2007; Muregi and Ishih, 2010; Shaveta et al., 2016), (ii) confirmation of role of IHB on (Z)-configuration of THIs, (iii) identify a readily available starting materials, environmentally benevolent catalyst and solvent for the above strategy, (iv) apply the established procedure for the preparation of a library of targeted THIs and (v) investigation of electrochemical properties of THIs.

In this article, a practical, straightforward, and green FCDR strategy for the synthesis of (Z)-THIs (6) has been demonstrated with high stereospecificity via an IHB directed approach, involving the reaction of methyl ketones (1), NBS (2), isatins (4) and thiosemicarbazide (5) in the presence of environmentally benevolent HClO₄-SiO₂ in ethanol at reflux temperature for 45–60 min in a single-step operation (Scheme 1). This domino reaction is enabled by the *in situ* formation of α-bromoketone intermediate (3) followed by its reaction with isatins (4) and thiosemicarbazide (5). The absolute structure of the compound, (Z)-3-(2-(4-(4-bromophenyl)thiazol-2-yl)hydrazono)indolin-2-one (6e) has been confirmed by single-crystal XRD. The role of IHB on (Z)-configuration of the synthesized (Z)-THIs (6) is proved by using a single crystal XRD and ¹H NMR studies. Further, the electrochemical properties of (Z)-THIs (6) have been investigated.

2. Materials and methods

2.1. General information

The ¹H NMR spectra were recorded on a Bruker NMR spectrometer (500 MHz). HRMS spectra were acquired using a Micromass ESI-TOF MS, and Bruker micrOTOF-QII mass spectrometer. LC-MS spectra were performed on Shimadzu-LCMS-2010 A mass spectrometer. XRD data was obtained by using a Rigaku Miniflex 600 diffractometer. The absorbance spectra were taken using a Shimadzu model UV-3100 UV-Visible spectrophotometer. FT-IR spectra were recorded on Bruker Alpha II FT-IR spectrometer. CV experiments were performed on a CH Instruments Inc., USA, CHI6002E equipped with a three-electrode assembly. The MR-Vis + instrument was used to determine the melting points (Labindia). TLC was used to monitor the progress of the reactions

Table 1Screen and optimization of the reaction conditions for **3a** and **6a**.^a

Entry	Catalyst (15 mg)	Solvent (7 mL)	Temp. (°C)	3a		6a	
				Time (min)	Yield ^b	Time (min)	Yield ^c
1 ^d	–	H ₂ O	RT	180	–	180	–
2	Al ₂ O ₃	H ₂ O	Reflux	30	–	90	–
		Acetone	RT	90	–	90	–
		Isopropanol	Reflux	30	10	90	–
		EtOH	Reflux	30	20	90	–
3	TiO ₂	H ₂ O	Reflux	90	–	90	–
		Acetone	RT	90	–	90	–
		Isopropanol	Reflux	30	–	90	–
		EtOH	Reflux	40	–	90	–
4	SiO ₂	H ₂ O	Reflux	90	–	90	–
		Acetone	RT	90	–	90	–
		Isopropanol	Reflux	30	50	90	25
		EtOH	Reflux	30	65	90	30
5 ^e	HClO ₄	H ₂ O	Reflux	90	–	90	–
		Acetone	RT	90	–	90	–
		Isopropanol	Reflux	90	35	90	25
		EtOH	Reflux	90	55	90	50
6	HClO ₄ –Al ₂ O ₃	H ₂ O	Reflux	90	–	90	–
		Acetone	RT	90	–	90	–
		Isopropanol	Reflux	30	30	90	25
		EtOH	Reflux	30	40	90	30
7	HClO ₄ –TiO ₂	H ₂ O	Reflux	90	–	90	–
		Acetone	RT	90	–	90	–
		Isopropanol	Reflux	30	30	90	25
		EtOH	Reflux	30	40	90	35
8	HClO ₄ –SiO ₂	H ₂ O	Reflux	90	–	90	–
		Acetone	RT	90	–	90	–
		Isopropanol	Reflux	30	75	90	55
		EtOH	Reflux	30	85	90	80

^a Reagents and conditions: 3-acetylcoumarin (**1a**) (5.0 mmol), NBS (**2**) (6.0 mmol), Isatin (**4a**) (5.0 mmol), thiosemicarbazide (**5**) (5.0 mmol), catalyst, solvent (7 mL), temperature (RT/Reflux).

^b Isolated yields of **3a**.

^c Isolated yields of **6a**.

^d Reaction performed in the absence of catalyst.

^e HClO₄ (2.5 mol%).

on 0.25 mm Merck silica gel plates, and the spots were visualized under UV light. All solvents and chemicals used in this study were procured from Merck, Sigma Aldrich, and Acros Organics Ltd, and used without further purification.

2.2. HClO₄–SiO₂ catalyzed synthesis of (Z)-thiazolyldiazonoindolin-2-ones (**6**)

The mixture of methylketones (**1**) (5.0 mmol), *N*-bromosuccinimide (**2**) (6.0 mmol), and HClO₄–SiO₂ (25 mg) in ethanol (7.0 mL) was stirred at the reflux temperature for 25–30 min. The formation of α-bromo ketones (**3**) was confirmed using TLC and ¹H NMR analyses. The reaction mass was cooled down to RT and then added isatins (**4**) (5.0 mmol) and thiosemicarbazide (**5**) (5.0 mmol) to the reaction mixture and stirred for 20–30 min under reflux temperature. After completion of the reaction (as monitored by TLC), 5 mL of water and 5 mL of ethyl acetate was added to the crude reaction mixture and the catalyst was separated via filtration. The ethyl acetate layer was separated, dried with anhydrous Na₂SO₄, solvent was removed and the formed product (**6**) was dried. The product (**6**) obtained was recrystallized from hot ethanol.

2.3. Gram-scale (15 g) synthesis of the (Z)-3-(2-(4-(4-bromophenyl)thiazol-2-yl)hydrazono)indolin-2-one (**6e**)

A mixture of *p*-bromoacetophenone (**1e**) (0.0754 mol), *N*-bromosuccinimide (**2**) (0.0854 mol) and HClO₄–SiO₂ (300 mg) in ethanol (75 mL) was stirred at reflux temperature for 1 h. The formation of 2-bromo-1-(4-bromophenyl)ethan-1-one (**3e**) was determined using TLC. After cooling the reaction to room temperature, isatins (**4**) (0.0754 mol) and thiosemicarbazide (**5**) (0.0754 mol) were added to the reaction mixture, and stirred for 45 min at reflux temperature. After completion of the reaction (as determined by TLC), 50 mL of water and 50 mL of ethyl acetate was added to the crude reaction mixture and the catalyst was separated via filtration. The ethyl acetate layer was separated, dried with anhydrous Na₂SO₄, solvent was removed and the formed product (**6e**) was dried. The product (**6e**) obtained was recrystallized from hot ethanol.

2.4. Preparation of HClO₄–SiO₂ catalyst

HClO₄ catalyst was prepared by following the reported procedure (Kumar et al., 2021; Gallo and Burtolos, 2018). 0.14 g (1.40 mmol) of perchloric acid (70% aqueous solution) was gradually added to the

Table 2Loading of catalyst for the improvement of yields of **3a** and **6a**.^a

Entry	HClO ₄ -SiO ₂ (mg)	Solvent (7 mL)	3a		6a	
			Time (min)	Yield ^b (%)	Time (min)	Yield ^c (%)
1	15	EtOH	30	85	90	80
2	20	EtOH	30	91	60	89
3	25	EtOH	30	98	45	98
4	30	EtOH	30	98	45	98
5	35	EtOH	30	98	45	98
6	40	EtOH	30	98	45	98

^a Reagents and conditions: 3-acetylcoumarin (**1a**) (5.0 mmol), NBS (**2**) (6.0 mmol), Isatin (**4a**) (5.0 mmol), thiosemicarbazide (**5**) (5.0 mmol) in EtOH (7 mL) at reflux temperature.

^b Isolated yields of **3a**.

^c Isolated yields of **6a**.

suspension of 10 g of SiO₂ (230–400 mesh) in 20 mL of EtOAc. At room temperature, the slurry was stirred for 30 min, and the solvent was removed under vacuum. The resulting residue was heated at 100 °C for 3 h in a vacuum chamber to obtain a free-flowing powder of HClO₄-SiO₂ catalyst (0.14 mmol/g). The present procedure for the preparation of HClO₄-SiO₂ catalyst has been compared with the previously reported procedures (Chakraborti and Gulhane, 2003; Khatik et al., 2007; Wu et al., 2010; Bandgar et al., 2010; Nasr-Esfahani et al., 2011; Chittimalla et al., 2014; Gallo and Burtolos, 2018; Kumar et al., 2021) (See SI, Table S1). As can be seen from the table, it is clear that the present procedure has relatively better features than most of the reported procedures, use of a low volume of environmentally benign ethyl acetate as a solvent, shorter reaction time, low concentration of HClO₄ (0.14 mmol/g), etc.

3. Results and discussion

A model reaction was conducted for synthesizing THIs (**6**) as illustrated in Scheme 1. For this purpose, 3-acetylcoumarin (**1a**), NBS (**2**), isatin (**4a**), and thiosemicarbazide (**5**) were selected as model substrates for the synthesis of **6a** via α -bromination followed by subsequent

heterocyclization and condensation. When the reaction was conducted in the absence of catalyst in H₂O at room temperature, the formation of **6a** was not observed (entry 1, Table 1). Next, the above reaction was conducted individually in the presence of various catalysts like Al₂O₃, TiO₂, SiO₂, HClO₄, HClO₄-Al₂O₃, HClO₄-TiO₂, and HClO₄-SiO₂ in different green solvents at RT/reflux temperature (entries 2–8, Table 1). After the examination of different catalysts in different solvents for above-said reaction, the following important observations were noticed, i) in the presence of silica (15 mg) in ethanol at reflux temperature α -bromination proceeds smoothly and produced **3a** in moderate isolated yields (65%) but the obtained **3a** was transformed into the desired product **6a** in very low yield (30%) (entry 4, Table 1); ii) similarly, in the presence of HClO₄ under the same reaction conditions **3a** was obtained in moderate yields (55%), and the formed **3a** was almost converted into the corresponding product **6a** (50%) (entry 5, Table 1); iii) interestingly, HClO₄-SiO₂ (15 mg) afforded a good yields of **3a** (85% in 30 min) and **6a** (80% in 90 min) in ethanol at reflux temperature (entry 8, Table 1) and iv) the other catalysts in different solvents provided **3a** and **6a** in lower to moderate yields at reflux temperature (entries 2, 3, 6 & 7, Table 1).

The amount of loaded HClO₄-SiO₂ (15, 20, 25, 30, 35, and 40 mg) was varied to optimize **6a** yield (Table 2). It was found in this study that 25 mg of HClO₄-SiO₂ provided the highest yield of **3a** (98%) in 30 min and **6a** (98%) in 45 min at reflux temperature in ethanol (entry 3, Table 2). The yields of the product remained the same despite an increase in the amount of HClO₄-SiO₂ (entries 4–6, Table 2). It is concluded that the said 4-component domino reaction proceeded smoothly with 25 mg of HClO₄-SiO₂ in ethanol and provided the maximum isolated yield of target product **6a**.

It is important to mention that the structure of the *in situ* generated 3-(bromoacetyl)coumarin (**3a**) (separated by preparative TLC) has been proved by ¹H NMR analysis (Fig. 1). From the ¹H NMR spectrum (CDCl₃ and a few drops of DMSO-*d*₆), it was noticed that the methylene protons (CH₂-Br) of **3a** appeared as a singlet at δ 4.77 ppm, indicating the formation of **3a** from 3-acetylcoumarin (**1a**) and NBS (**2**) via α -bromination.

With this established procedure in hand, the generality of this FCDR has been investigated by using various methylketones (**1b-j**), NBS (**2**),

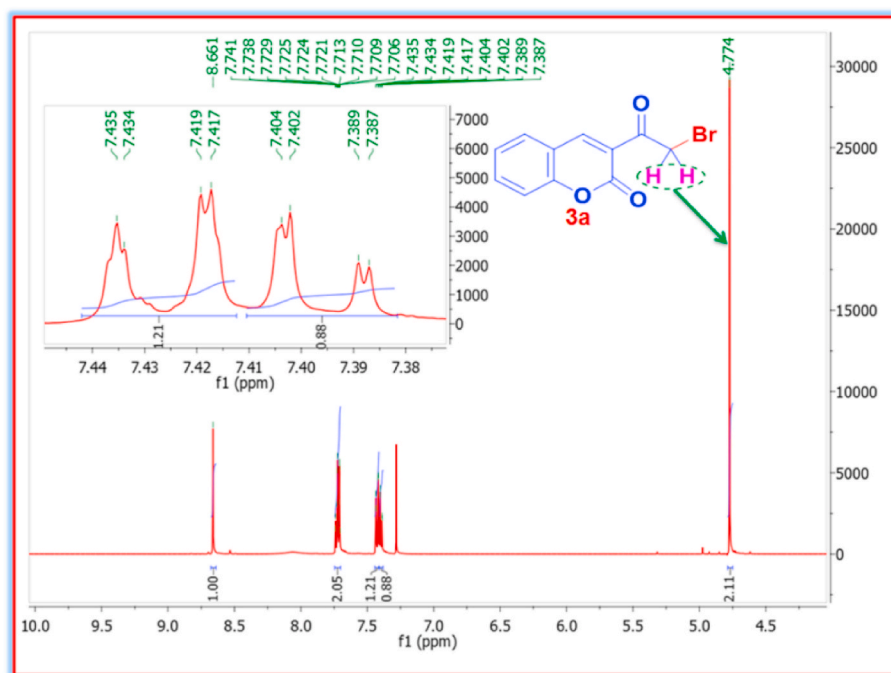
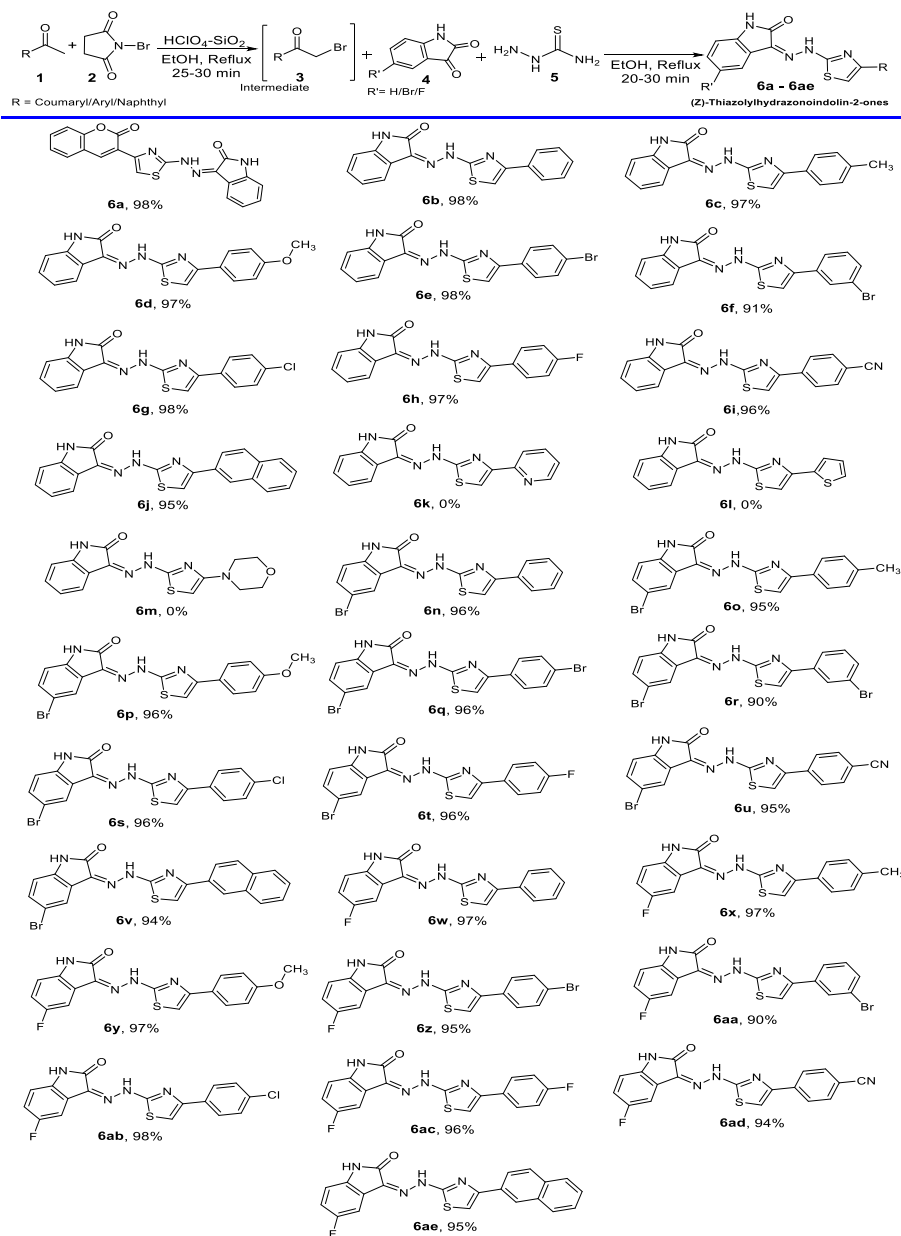


Fig. 1. ¹H NMR spectrum of 3-(bromoacetyl)coumarin (**3a**).

Table 3
Synthesis of a library of (Z)-thiazolylhydrazonoindolin-2-ones (6).



^aReagents and conditions: Methylketones (1) (5.0 mmol), NBS (2) (6.0 mmol), Isatins (4) (5.0 mmol), thiosemicarbazide (5) (5.0 mmol), $\text{HClO}_4\text{-SiO}_2$, EtOH, Reflux.

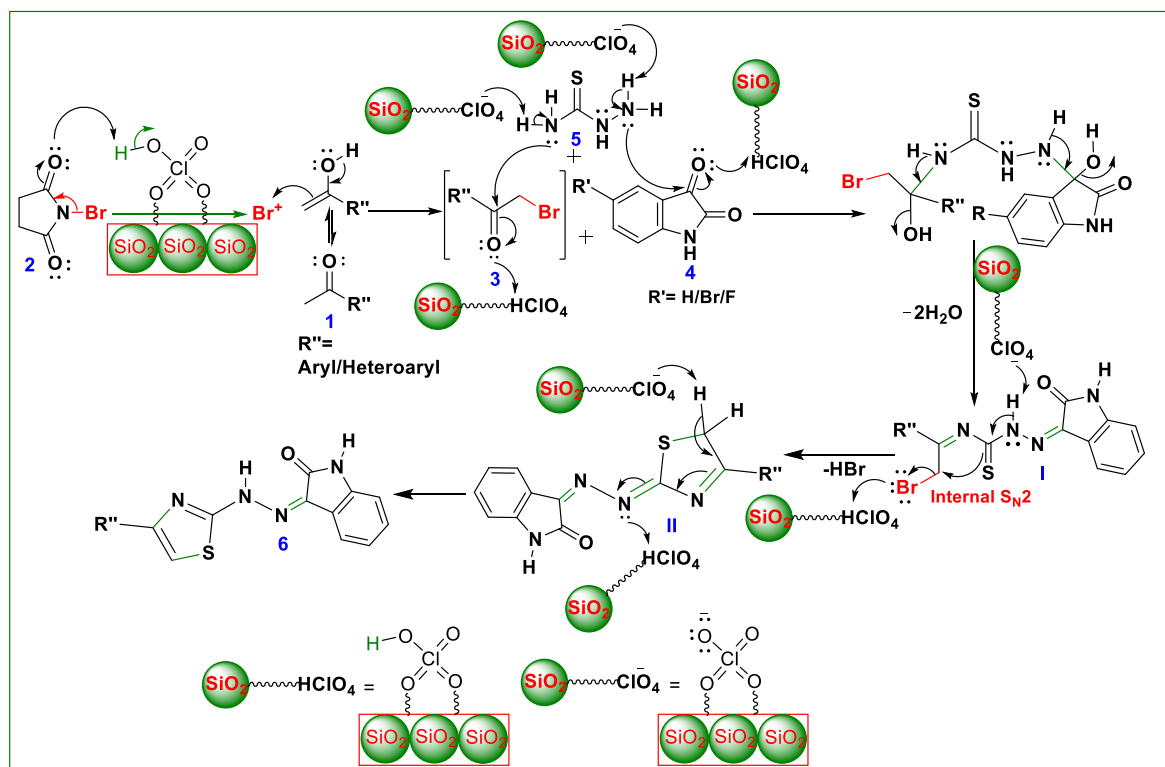
substituted isatins (4a-c), and thiosemicarbazide (5). The obtained results are summarized in Table 3. Isatin (4a) exhibited good reactivity with *in situ* generated phenacyl bromide (3b) (obtained via α -bromination of acetophenone) and thiosemicarbazide (5), generating the corresponding THI product (6b) in excellent yield (95%). Similarly, acetophenones with activating groups (*p*-Me (1c), *p*-OMe (1d)), and deactivating groups (*p*-Br (1e), *m*-Br (1f), *p*-Cl (1g), *p*-F (1h) and *p*-CN (1i)) (1b-i) at *para/meta* positions on the phenyl ring afforded the desired THI products (6b-i) in good to excellent yields (91–98%). The absolute structure of compound 6e has been confirmed by X-ray crystallography (Fig. 3). Employing 2-acetylnaphthalene (1j) also afforded the corresponding THI product (6j) with an excellent yield (95%). It was also observed that the 2-acetylpyridine (1k), 2-acetyl thiophene (1l), and 4-acetylmorpholine (1m) did not provide the corresponding products 6k, 6l, and 6m. This indicates no formation of α -bromoketones of

compounds 1k, 1l, and 1m, respectively.

To expand the scope of our method, substituted isatins such as 5-bromoisatin (4b) and 5-fluoro isatin (4c) were also examined under the optimized reaction conditions. Thus, 5-bromo isatin (4b) and 5-fluoro isatin (4c) also showed good reactivity with a variety of *in situ* generated α -bromo ketones (3b-j) (obtained on α -bromination of respective methyl ketones) and thiosemicarbazide (5) afforded the desired THI products (6n-6v) and (6w-6ae) were ranged from 90 to 98%. From these results, it was observed that the yields of THIs (6) depend on the percentage of *in situ* formed α -bromomethyl ketones (3).

3.1. Recyclability

The catalyst recyclability test was performed for the synthesis of 6e by using 4-bromoacetophenone (1e), NBS (2), isatin (4a), and



Scheme 2. A possible mechanism for the synthesis of (Z)-THIs (6) catalyzed by $\text{HClO}_4\text{-SiO}_2$.

thiosemicarbazide (5) under the established procedure. After completion of reaction, the catalyst was recovered using simple filtration, washed with 1.0 mL of ethyl acetate, and dried. Further, the recovered catalyst was reused successfully for seven times, and the obtained yields of **6e** were 98, 98, 98, 97, 96, 96, and 96% from 1st to 7th cycle, respectively. The study reveals that recycled catalysts perform similarly to fresh catalysts in terms of catalytic efficiency.

3.1.1. Hot filtration test

The heterogeneity of $\text{SiO}_2\text{-HClO}_4$ was investigated by a hot filtration test for the synthesis of (Z)-3-(2-(4-(4-bromophenyl)thiazol-2-yl)hydrazono)indolin-2-one (**6e**) using 4-bromoacetophenone (**1e**), NBS (**2**), isatin (**4a**), and thiosemicarbazide (**5**) under the optimized reaction conditions. The reaction was refluxed for 35 min after which the catalyst was separated by filtration under hot conditions, and the obtained yield of **6e** was 55%. The reaction was then continued in the absence of catalyst under similar reaction conditions for another 60 min. The study revealed that the reaction didn't proceed further without catalyst. Hence, it is concluded that the catalyst is completely heterogeneous in nature.

3.2. Scalability

In order to check the scalability of the optimized green method, the reaction of 4-bromoacetophenone (**1e**), NBS (**2**), isatin (**4a**), and thiosemicarbazide (**5**) catalyzed by $\text{HClO}_4\text{-SiO}_2$ in different gram-scale reactions (i.e., 5.0 and 10.0 15.0 g) have been carried out. The obtained yields of product **6e** were 98, 97, and 97%, respectively. Hence, the developed FCDR strategy is most suitable for the scale-up process of (Z)-thiazolylhydrazoneindolin-2-ones (**6**).

3.3. Mechanism

A possible mechanism for the formation of (**6**) catalyzed by $\text{HClO}_4\text{-SiO}_2$ is depicted in Scheme 2. Firstly, methyl ketones (**1**) undergo

regioselective α -bromination with NBS (**2**) in the presence of $\text{HClO}_4\text{-SiO}_2$ to form α -bromoketones (**3**). Next, two amino groups of thiosemicarbazide (**5**) react with the keto groups of both isatin (**4**) and α -bromo ketones (**3**), which are activated by $\text{HClO}_4\text{-SiO}_2$ to obtain diimine (**I**). Later, it undergoes $\text{S}_\text{N}2$ displacement of the bromide ion to generate an intermediate (**II**), subsequently, a proton shift to form (Z)-THIs (**6**).

3.4. Confirmation of role of IHB on Z-specific synthesis of THIs (**6**) by X-ray crystallography and ^1H NMR study

The molecular structure of compound **6e** (Fig. 2a) was analyzed using single-crystal X-ray analysis. Compound **6e** has a single monoclinic crystal with a $\text{P}_{21/\text{c}}$ space group and one molecule in an asymmetric unit ($Z = 4$, and $Z' = 0$). The absolute configuration of compound **6e** was confirmed as Z-configuration by single-crystal XRD characterization. The stability of this Z-isomer was due to the IHB between the oxygen atom of isatin carbonyl group and NH proton of hydrazone group, $\text{C}=\text{O}\cdots\text{H}-\text{N}$ ($\text{O}\cdots\text{H} = 2.032 \text{ \AA}$, $\text{O}\cdots\text{H}-\text{N} = 120^\circ$) in a stable cyclic six-membered ring transition state (Fig. 2b). The distance of IHB ($2.032 \text{ \AA}$) is within the range of reported IHBs (Lee et al., 2006; Kang, 2000; Gilli et al., 2000). This is further supported by ^1H NMR analysis. From the ^1H NMR spectra of (Z)-THIs (**6**), the hydrazone NH proton of (Z)-THIs (**6**) appeared as a singlet ($\delta > 13$ in $\text{DMSO}-d_6$) in the downfield region, which indicates the presence of IHB in (Z)-THIs (**6**) (Fig. 2c).

3.5. Comparison

The merits of the present method were explored by comparing it with the earlier reports on the synthesis of (Z)-THIs (**6**) (Table 4). As can be seen, it was evident that the developed FCDR strategy has several advantages over the reported protocols, mainly the avoidance of highly lachrymatory α -bromoketones as starting materials, shorter reaction times, better yields, broad substrate scope, and scalability. Further, the used catalyst is environmentally benign, inexpensive, recoverable, and

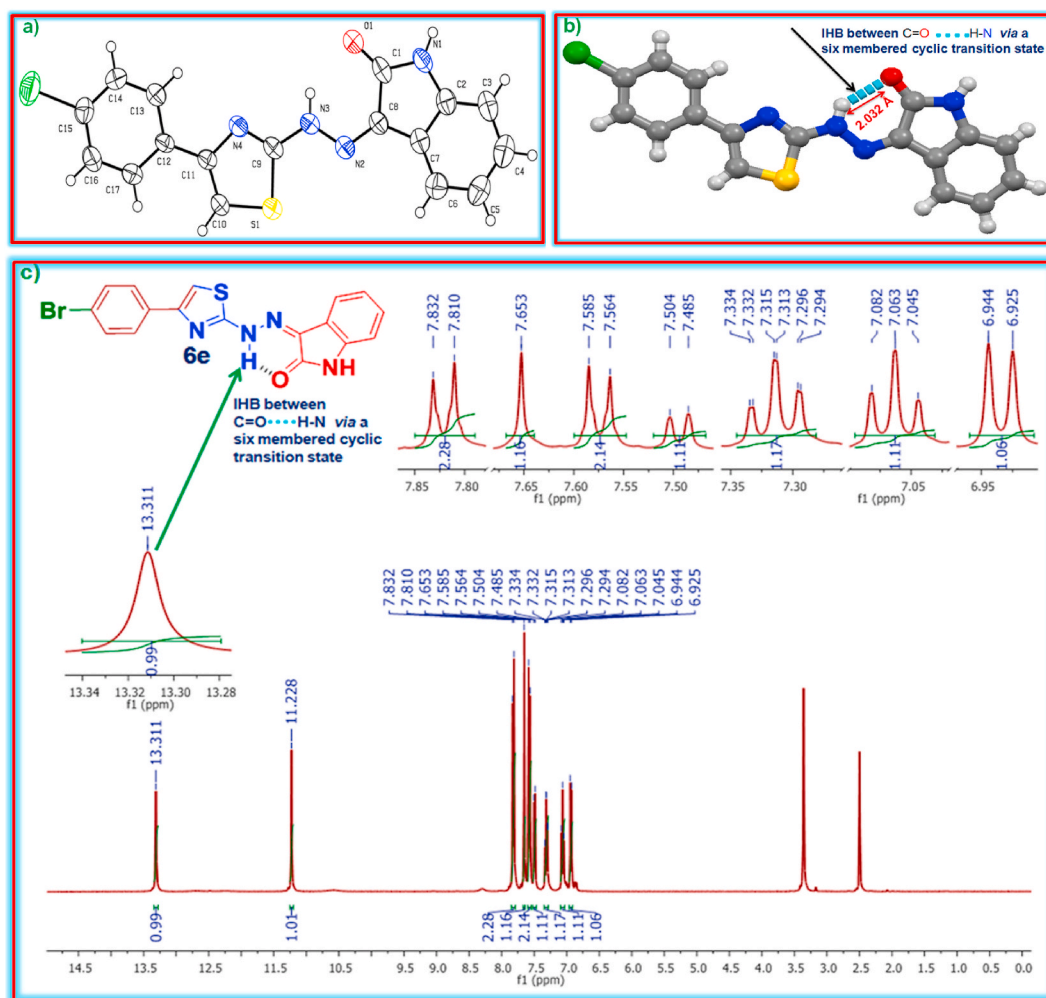


Fig. 2. a) ORTEP diagram of **6e** with 50% probability; b) IHB between the oxygen atom of isatin carbonyl group, and NH proton of hydrazone (dotted lines show IHB); c) ^1H NMR spectrum of **6e**.

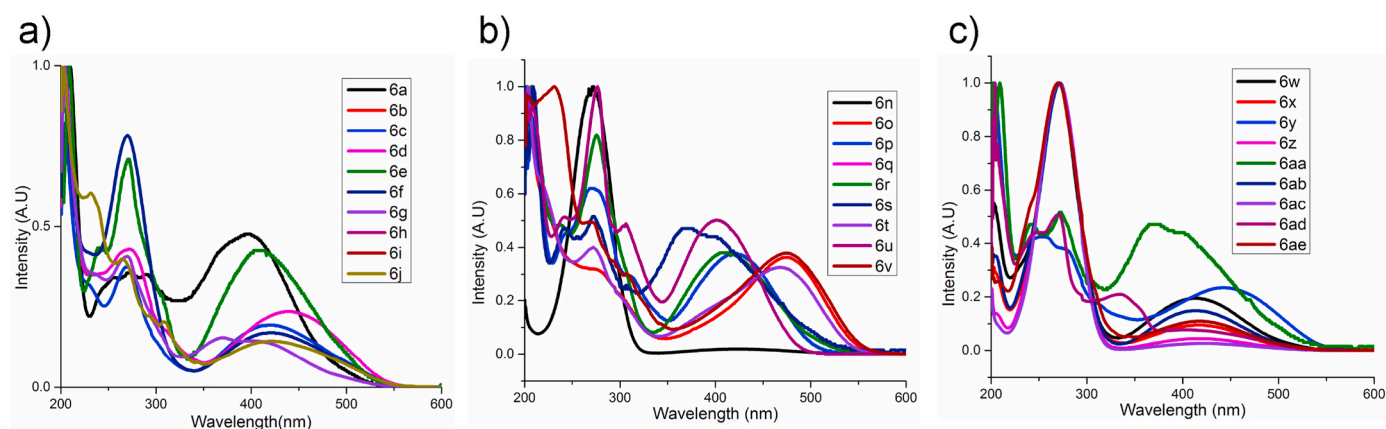


Fig. 3. Absorption spectra of THIIs a) 6a-6j b) 6n-6v and c) 6w-6ae in CH_3OH (1.0×10^{-5} M).

reusable.

3.6. Electrochemical properties

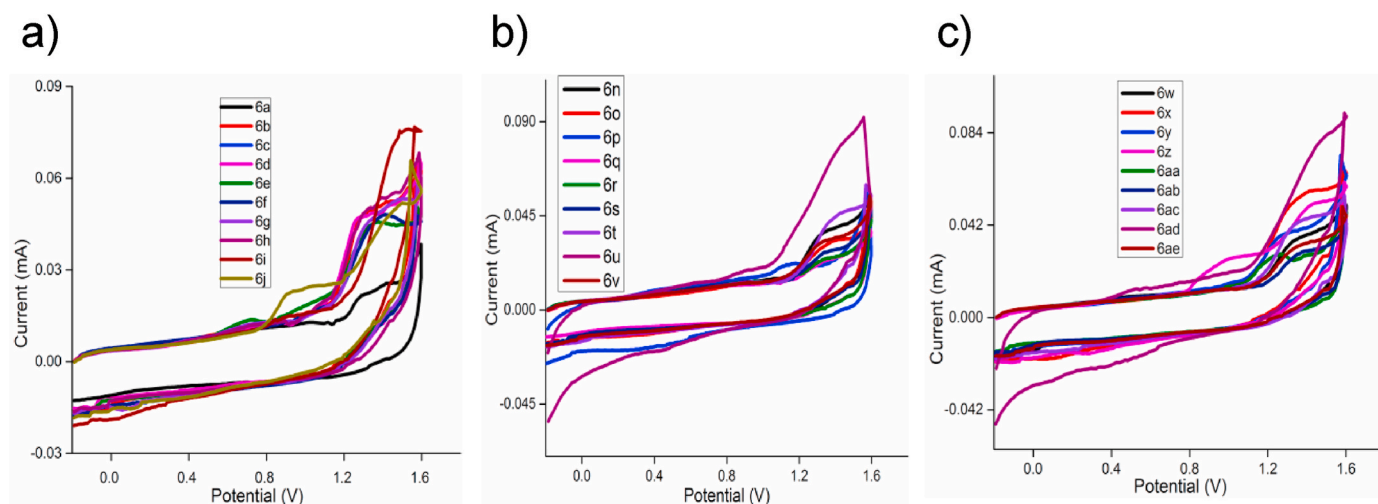
The hole transport, electron transport and ambipolar behaviour of THIIs (**6**) have been studied by cyclic voltammetry (CV). The CV measurements of THIIs (**6**) were carried out in CH_3OH solution (0.5 mM)

along with supporting electrolyte (Bu_4NPF_6 (0.1 M)). Ag/Ag^+ (satd. KCl), platinum disk, and platinum wire were employed as a reference electrode, working electrode and counter electrodes. All CV scans were performed at a scan rate of 100 mV/s in the potential range -0.2 to 1.6 V vs Ag/AgCl (sat. KCl). Fig. 4 illustrates cyclic voltammograms of THIIs (**6**). The HOMO and LUMO energy levels have been calculated (Kumar et al., 2020, 2021) and presented in Table 5.

Table 4

Comparing the results of the present method with previously reported methods on the synthesis of (Z)-THIs (6).

Starting materials	Catalyst/Solvent	Type of MCR	Time (min)	Temp.	Yield (%)	Scope of the method	Ref
Methyl ketones, NBS, isatins and thiosemicarbazide	HClO ₄ -SiO ₂ /EtOH	4-component	40–65	Reflux	90–98	28 examples	Present work
α-Bromoketones, isatins and thiosemicarbazide	CH ₃ COOH/EtOH	3-component	120–180	Reflux	88–91	08 examples	Navaneethgowda et al. (2022)
α-Bromoketones, isatins and thiosemicarbazide	CTAB/Glycerol	3-component	90–150	45 °C	86–95	05 examples	Tiwari et al. (2018)
α-Bromoketones, isatin and thiosemicarbazide	CH ₃ COOH/EtOH	3-component	180–360	Reflux	75–85	11 examples	Tewodros Birhanu and Rajeswar Rao (2014)

**Fig. 4.** Cyclic voltammograms of THIs a) 6a-6j b) 6n-6v and c) 6w-6ae in CH₃OH (1.0×10^{-5} M) at the scan rate of 100 mV/s.**Table 5**

The electrochemical data of THIs.

Compd.	λ_{edge} (nm)	$E_{\text{g}}^{\text{opt}}$ (eV) ^a	$E_{\text{onset}}^{\text{ox}}$	E_{HOMO} (eV) ^b	E_{LUMO} (eV) ^c
6a	492	2.52	1.195	-5.62	-3.10
6b	314	3.95	1.067	-5.49	-1.54
6c	551	2.25	1.044	-5.46	-3.21
6d	541	2.29	1.055	-5.48	-3.19
6e	523	2.37	0.856	-5.28	-2.91
6f	325	3.82	1.081	-5.50	-1.68
6g	323	3.84	1.032	-5.45	-1.61
6h	341	3.63	1.078	-5.50	-1.87
6i	288	4.30	1.180	-5.60	-1.30
6j	545	2.27	1.170	-5.59	-3.32
6n	317	3.91	1.159	-5.58	-1.67
6o	554	2.24	1.095	-5.52	-3.28
6p	499	2.48	1.023	-5.44	-2.96
6q	518	2.39	1.135	-5.55	-3.16
6r	535	2.32	1.090	-5.51	-3.19
6s	515	2.41	1.139	-5.56	-3.15
6t	551	2.25	1.099	-5.52	-3.27
6u	490	2.53	1.103	-5.52	-2.99
6v	552	2.25	1.107	-5.55	-3.30
6w	329	3.77	1.146	-5.57	-1.80
6x	317	3.91	1.102	-5.52	-1.61
6y	546	2.27	1.046	-5.47	-3.20
6z	312	3.97	1.185	-5.61	-1.64
6aa	516	2.40	1.081	-5.50	-3.10
6ab	321	3.86	1.155	-5.58	-1.72
6ac	314	3.95	1.138	-5.56	-1.61
6ad	296	4.19	1.116	-5.54	-1.35
6ae	323	3.84	1.126	-5.55	-1.71

^a $E_{\text{g}}^{\text{opt}} = 1240/\lambda_{\text{edge}}$.^b $E_{\text{HOMO}} = -[E_{\text{ox}}(\text{onset}) - E_{\text{FOC}} + 4.8]$ [$E_{\text{FOC}} = 0.38$ eV (FOC/FOC⁺ vs. Ag/AgCl)].^c $E_{\text{LUMO}} = E_{\text{HOMO}} + E_{\text{g}}^{\text{opt}}$.

The majority of the cyclic voltammograms of THIs (6) were quasi-reversible in nature (Fig. 4). Therefore, the HOMO levels were determined using onset potentials of the oxidation wave, and the LUMO levels were determined using HOMO levels and the optical band gaps ($E_{\text{g}}^{\text{opt}}$) (calculated by the onset of UV absorption ranging from 2.24 to 4.30 eV (Fig. 3 and Table 5). As stated in the previous literature reports, the APMs have HOMO, and LUMO energy levels that must be ≥ -5.6 eV and ~ -3.15 eV (Zhou et al., 2014). As can be seen from Table 5, the HOMO and LUMO energy levels of THIs, 6a, 6c, 6d, 6j, 6o-6v, 6y & 6aa ranged from -5.44 to -5.62 eV and -2.96 to -3.32 eV, respectively. The obtained results were comparable to the previously reported APMs (Thompson and Frechet, 2008; Liang et al., 2011; Li et al., 2012; Wang et al., 2013). The HOMO levels of other THIs, 6b, 6e-6i, 6n, 6w, 6x, 6z, and 6ab-6ae ranged from -5.61 eV to -5.28 eV, and these energy levels were in good agreement with the most commonly utilized HTMs (Shahnawaz et al., 2019; Jou et al., 2015; Chen et al., 2014). Hence, the THIs are promising ambipolar and HTMs to fabricate optoelectronic devices. Further, the HOMO energy levels of all the studied THIs (6) were less than the threshold for air oxidation (~ -5.2 eV), indicating the synthesized THIs (6) have good stability under ambient conditions (Viswanathan et al., 2017). The study revealed that the substituents on isatin and phenyl ring of THIs and intramolecular hydrogen bonding strongly influence the electrochemical properties of THI compounds (6).

4. Conclusions

An environmentally accountable HClO₄-SiO₂ catalyzed FCDR strategy has been established for the synthesis of (Z)-THIs (6) through an IHB directed approach. Single-crystal XRD and ¹HNMR analyses confirm the role of IHB on Z-configuration of THIs. Formation of C-Br, C-N, C-S, and C=N bonds through a one-pot process, avoidance of highly lachrymatory α-bromoketones as starting materials, simplicity of workup procedure,

good functional group tolerance, high yields, shorter reaction times, gram scale feasibility, products free from chromatographic purification, and reuse of catalyst are the important features of this FCDDR approach. Hence, the developed FCDDR strategy is a convenient and greener synthetic route for synthesizing a library of (Z)-THIs. The limitation of the present method is that the catalyst did not work for the heterocyclic methyl ketones such as 2-acetylpyridine (**1k**), 2-acetyl thiophene (**1l**), and 4-acetylmorpholine (**1m**). The electrochemical study indicates that HOMO and LUMO levels of **6a**, **6c**, **6d**, **6j**, **6o-6v**, **6y**, and **6aa** are comparable with the most commonly used ambipolar organic materials, and the HOMO energy levels of **6b**, **6e-6i**, **6n**, **6w**, **6x**, **6z** and **6ab-6ae** are similar with the reported HTMs. Hence, the synthesized (Z)-THIs (**6**) could be utilized as ambipolar and hole-transporting materials to fabricate a wide range of optoelectronic devices. Luminescence studies of (Z)-THIs (**6**) are in progress. Furthermore, the prepared synthetic precursors (**6**) might be served as starting point for the development of new electroluminescence materials through structural elaboration and functional group transformations.

Credit author statement

Sultana Shaik: Investigation, Methodology, Formal analysis, Writing - Original Draft, Rama Mohana Reddy Sirigireddy: Methodology, Kumar Godugu: Methodology, Formal analysis, Venkatramu Vemula: Reviewing & Validation, Raghava Reddy Kakarla: Formal analysis, Supervision, Writing - Reviewing & Editing, Ekambaram Balaraman: Writing - Reviewing & Editing, Chinna Gangi Reddy Nallagonda: Conceptualization, Formal analysis, Funding acquisition, Supervision, Writing - Reviewing & Editing, Tejraj M. Aminabhavi: Supervision, Writing - Reviewing & Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix. ASupplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chemosphere.2022.136667>.

Appendix. BList of abbreviations

Abbreviation Compound

MCDRs Multi-component domino reactions
FCDDR 4-component domino reaction

THIs	Thiazolyldiazonoindolin-2-ones
NBS	N-bromosuccinimide
IHB	Intramolecular hydrogen bond
DOS	Diversity-oriented synthesis
OLEDs	Organic light emitting diodes
OPVs	Organic photovoltaics
OFETs	Organic field effect transistors
HIL/HTL	Hole-injection/transport layer
HTMs	Hole transport materials
ETMs	Electron transport materials
APMs	Ambipolar materials
EML	Emissive layer
EIL/ETL	Electron-injection/transport layer
HOMO	Highest occupied molecular orbital
LUMO	Lowest unoccupied molecular orbital
CV	Cyclic voltammetry
CTAB	Cetyltrimethylammonium bromide
MNPs	Magnetic nanoparticles
SBA-15	Santa Barbara Amorphous-15
MCM-41	Mobil composition of matter No. 41
LDH	Layered double hydroxide
HEPES	N-(2-Hydroxyethyl)piperazine-N'-(2-ethanesulfonic acid)
NPs	Nanoparticles
HPW	Heteropoly acid
Cu(OTf) ₂	Copper(II) trifluoromethanesulfonate
EtOH	Ethanol
¹ H NMR	Proton nuclear magnetic resonance
HRMS	High-resolution mass spectrometry
ESI-TOF-MS	Electrospray ionization time-of-flight mass spectrometry
LCMS	Liquid chromatography-mass spectrometry
TLC	Thin-layer chromatography
XRD	X-Ray diffraction

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